

A hybrid QSVM-QNN approach for breast cancer diagnosis using quantum feature maps

Vikesh Yadav¹, Savita Kumari Sheoran¹ and Rakesh Kumar Sheoran²

¹ Department of Computer Science & Engineering, Indira Gandhi University Meerpur, Rewari, Haryana, India

² SIASTE, Gurugram, Haryana, India

ABSTRACT

Quantum assisted machine learning (QML) have the potential to outperform the classical Machine Learning models by utilizing the quantum feature maps embeddings and variational circuits. In this study, a Hybrid Quantum Support Vector Machine (QSVM) and Quantum Neural Network (QNN) is proposed for supervised classification applied to the Breast Cancer Wisconsin Dataset. The QSVM component employs fidelity quantum kernels with three distinct feature maps—Z, ZZ and Pauli—to transform classical data into quantum states. The QNN is implemented using a variational quantum classifier based on the EstimatorQNN, which enables quantum-trainable parameters for enhanced feature extraction. The proposed hybrid methodology is evaluated using seven principal features, and experiments are conducted on IBM's state vector Qiskit Aer local simulator. Performance is assessed based on accuracy, recall, precision, F1-score, and confusion matrix. Additionally, statistical significance is evaluated using paired t test and F test to compare the performance of different quantum feature maps and to validate improvements made in the Hybrid Model. The results demonstrate that the hybrid QSVM/QNN approach improves classification performance not only in terms of accuracy by achieving above 90% accuracy but also by reducing computational cost. This work contributes to the advancement of quantum enhanced machine learning (ML) algorithms by demonstrating the hybrid quantum-classical approach in structured data classification.

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Corresponding author

Vikesh Yadav,
vikeshyadav1994@gmail.com

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INTRODUCTION

Recently, the regime of machine learning (ML) has achieved significant advancements, primarily because of the accessibility of extensive datasets and the emergence of robust computer platforms. It is a foundational concept in the arena of artificial intelligence. The goal is to develop systems capable of autonomously learning from data, enabling decision-making with little dependence on human interference. It has become a central topic of investigation in both scholarly and industrial sectors, with significant applications in critical areas such as image recognition, public safety, healthcare, intelligent transportation, and natural language processing.

ML has advanced significantly across multiple regimes, such as industry ([Datta & Davim, 2022](#); [Li, 2019](#)), natural language processing ([Dwivedi & Dwivedi, 2022](#); [Chen et al., 2024](#)), healthcare ([Javaid et al., 2022](#); [Miceli & McGuigan, 2018](#)), pattern recognition ([Singh, 2023](#); [Fernandez-Carames & Fraga-Lamas, 2020](#)), medicine, education ([Ersozlu, Taheri & Koch, 2024](#); [Esposito, Uehara & Spanias, 2022](#)) and data mining ([Li, 2019](#); [Masum et al., 2023](#)). It empowers computers with decision-making and logical thinking capabilities to address intricate and unprecedented challenges. In machine learning approaches, the quality of data is directly correlated with the model's extent of learning and generalisation. Consequently, the majority of investigations have employed feature engineering and extraction techniques. It has been used to replicate human brain processes to develop models capable of extracting critical aspects from raw and processed data. Machine learning predominantly uses algebraic equations for processing information, utilising array data as input. Developments in such technologies have hampered classical computing constraints due to its bit-based designs and conventional methodologies. According to the reports of the International Data Corporation (IDC), there is an increase in global data generation from 33ZB in 2018 to 175ZB by 2025, with an average daily output of approximately 491EB ([Javaid et al., 2022](#)). So, there will be an enduring urge to enhance classical systems' computational speed and storage capability. However, Classical computers encounter numerous obstacles, including performance constraints arising from data proliferation, and processing high-dimensional complex data. Thus, These limitations have necessitated the investigation of novel computing paradigms, with quantum computing commonly regarded as capable of transcending this hardware constraint ([Mandviwalla, Ohshiro & Ji, 2019](#); [Li, 2019](#)).

Quantum computing, however, can surmount these constraints by utilising quantum bits or qubits. It is an emerging discipline that amalgamates mathematics, computer mathematics, computer science, and physics for the development of quantum computers (QCs). Unlike bits, qubits can represent values of 0, 1, or a superposition of both, hence facilitating the concurrent processing of extensive matrix data. Quantum computing is founded on the principles of quantum physics, which regulates the behaviour of matter at molecular and subatomic levels, asserting that particles can display wave-like and particle-like properties ([Miceli & McGuigan, 2018](#); [Bhatia & Ramkumar, 2020](#)). In contrast to classical computers, quantum computers can perform specific operations much more quickly due to their ability to exist in multiple states simultaneously through superposition, facilitated by quantum parallelism. Performing routine activities utilising quantum computers is rarely quicker or slower than that achieved with traditional computing. Nonetheless, the duration that classical computers require to resolve particular issues is yet to be established. Quantum computers can efficiently resolve analogous problems swiftly and with less storage requirements by utilising particular quantum algorithms. Nonetheless, this does not suggest that all responsibilities should be delegated to QCs. Quantum computing possesses numerous applications in cryptography ([Fernandez-Carames & Fraga-Lamas, 2020](#); [Omar & Abd El-Hafeez, 2025](#)), healthcare ([Agarwal & Alam, 2022](#); [Esposito, Uehara & Spanias, 2022](#)), and artificial intelligence

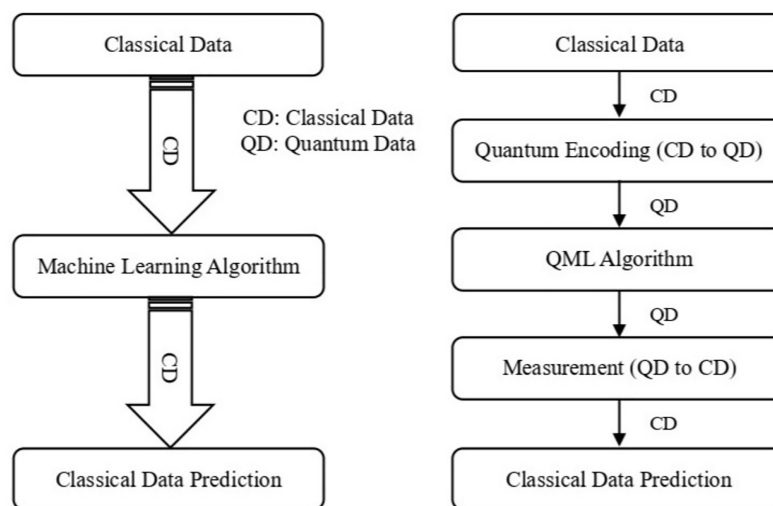


Figure 1 Classical vs quantum machine learning.

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(Abdelgaber & Nikolopoulos, 2020; Liu et al., 2025). Applying machine learning algorithms grounded in quantum computing principles has established a novel domain known as quantum machine learning (QML) (Akrom, 2024; Masum et al., 2023), markedly enhancing classical machine learning applications. Moreover, quantum computers may address challenges that exceed the capacities of the most advanced traditional supercomputers. To improve the speed and efficiency of machine learning compared to classical computing, quantum machine learning investigates the design and implementation of quantum software. The comparison of Classical vs Quantum Machine Learning is shown in Fig. 1 (Elsedimy, Elhadidy & Abohashish, 2024; Omar & Abd El-Hafeez, 2025). QML implements quantum algorithms, potentially exceeding classical ML techniques in particular scenarios.

The primary contributions of this work include:

- This article introduces a hybrid quantum support vector machine (QSVM) and quantum neural network (QNN) approach for supervised classification, which is applied to the Breast Cancer Wisconsin dataset.
- QSVM component utilises Fidelity Quantum Kernels with three feature maps-Z Feature Map, ZZ Feature Map, and Pauli Feature Map- to encode classical data into a higher-dimensional quantum Hilbert space. The QNN component is implemented using a Variational Quantum Classifier (VQC) based on an Estimator Quantum Neural Network (EstimatorQNN), allowing trainable quantum parameters for enhanced feature extraction.
- The proposed algorithm, a hybrid QSVM→QNN methodology, is evaluated using seven principal features, with experiments conducted on state vector simulator in Qiskit Aer.
- Performance is evaluated using classification accuracy, precision, recall, and F1-score, confusion matrix, t test and F test demonstrating the effectiveness of hybrid quantum-classical models for structured data classification.

Table 1 Existing quantum assisted algorithms.

Algorithms	References
Quantum SVM	<i>Ruskanda et al. (2023), Singh et al. (2022), Kariya & Behera (2021), Slys et al. (2023), Wang (2023), Zhang et al. (2023), Abbas et al. (2021), Abuga (2023), Adeniyi & Kumar (2025), Arthur & Date (2022), Bischof et al. (2025), Dutta et al. (2025), Easom-McCaldin et al. (2022), Fan et al. (2023), Fan, Shi & Zhu (2024), Henderson et al. (2019), Hur, Kim & Park (2022), Innan et al. (2024), Agarwal & Alam (2022), Akpinar, Ismal & Oduncuoglu (2024), Akrom (2024), Dema, Arai & Horikawa (2020), Bishwas, Mani & Palade (2020), Delilbasic et al. (2021), Ding, Bao & Huang (2022), Elsedimy, Elhadidy & Abohashish (2024), Farooq et al. (2024), Innan et al. (2023), Orazi et al. (2024), Rebentrost, Mohseni & Lloyd (2014)</i>
Quantum neural network	<i>Orazi et al. (2024), Jeong et al. (2023), Jeswal & Chakraverty (2019), Kwak et al. (2021), Li, Lu & Deng (2022), Li et al. (2024), Matic et al. (2022), Oh, Choi & Kim (2009), Patel & Tiwari (2016), Pira & Ferrie (2024), Patel et al. (2019), Savvas, Lizarralde & Marsoit (2023), Sebastianelli et al. (2022), Shi et al. (2023), Tomal et al. (2024), Uehara et al. (2021), Wang, Richerme & Chen (2023), Zhao & Wang (2021), Zidan et al. (2019), Zlokapa, Neven & Lloyd (2021), Dang et al. (2018), Feng et al. (2023), Li et al. (2022), Ruan et al. (2017), Wang et al. (2019), Wiebe, Kapoor & Svore (2015), Glos, Krawiec & Zimborás (2022), Li, Song & Wang (2021), Simonetti & Perri (2022), Cerezo et al. (2025), Gordon et al. (2022), Kavitha & Kaulgud (2023), Abbas et al. (2021), Abuga (2023), Adeniyi & Kumar (2025), Arthur & Date (2022), Bischof et al. (2025), Dutta et al. (2025), Easom-McCaldin et al. (2022), Fan et al. (2023), Fan, Shi & Zhu (2024), Farhi & Neven (2018), Henderson et al. (2019)</i>
Quantum K nearest neighbor	<i>Kerenidis et al. (2019), Chen et al. (2020), Xiao et al. (2010), Sharma, Singh & Kumar (2025), Ventura & Martinez (1998), Schuld, Sinayskiy & Petruccione (2014), Dang et al. (2018), Feng et al. (2023), Li et al. (2022), Ruan et al. (2017), Wang et al. (2019), Wiebe, Kapoor & Svore (2015)</i>
Variational quantum classifier	<i>Arthur & Date (2022), Liu & Rebentrost (2018), Raubitzek & Mallinger (2023), Senokosov et al. (2023), Narottama, Mohamed & Aissa (2023), Glos, Krawiec & Zimborás (2022), Innan et al. (2023), Li, Song & Wang (2021), Simonetti & Perri (2022), Cerezo et al. (2025)</i>
Principal component	<i>Innan et al. (2023), Gordon et al. (2022)</i>
K means	<i>Dai et al. (2024), Bilal et al. (2025), Kwon et al. (2025), Maragkopoulos et al. (2025), Chen et al. (2020), Kavitha & Kaulgud (2023), Kerenidis et al. (2019), Xiao et al. (2010)</i>
Decision tree	<i>Jura & Udrescu (2025), Sharma, Singh & Kumar (2025)</i>

Table 2 Comparison of quantum vs classical algorithms.

Algorithm	Quantum ML	Classical ML
Support vector machine	$O(\log(n \cdot m))$	$O(m \cdot n)$
K-nearest neighbor	$O(\log(n \cdot m))$	$O(n \cdot m)$
Principle component analysis	$O(\log(d))$	$O(d)$
K means	$O(k \cdot \log(k \cdot m \cdot n))$	$O(m \cdot n)$ (<i>Mandviwalla, Ohshiro & Ji, 2019; Chen et al., 2024</i>)

The structure of this article is as follows: ‘Introduction’ describes the introduction of quantum machine learning, ‘Literature Review’ presents the existing work, ‘Methodology’ defines the methodology, and ‘Results and Discussion’ illustrates the results, followed by a conclusion.

LITERATURE REVIEW

By combining both the QSVM and QNN techniques leverages an opportunity to integrate the strengths of both models, presenting the robustness of quantum kernels with the adaptability of variational circuits. In this section a critical review of existing researches on QSVM, QNN, and hybrid quantum-classical methodologies, emphasising theoretical

advancements and empirical benchmark performance. Table 1 represents the existing quantum-assisted ML algorithms with their complexity in Table 2.

Quantum machine learning (QML) has gained prominence as an emerging field and a potential domain that utilises quantum computing concepts to improve classical machine learning techniques. Quantum Neural Networks and Quantum Support Vector Machines have garnered interest among diverse quantum methodologies for their prospective benefits in high-dimensional data classification and pattern identification. Conventional machine learning models frequently encounter challenges related to scalability and computational difficulty, especially when dealing with extensive datasets characterised by complicated feature interrelationships. The incorporation of quantum feature maps into neural networks has the potential to leverage quantum parallelism and entanglement for enhanced classification efficacy. At first, a Quantum Computational learning algorithm was proposed by Ventura & Martinez (1998), which Examined the possibility of quantum computation to provide a learning method. Delilbasic et al. (2021) presented a quantum support vector machine and considered remote sensing data classification for evaluation. Aimer et al. proposed utilising the overlap or fidelity between two quantum states $|a\rangle$ and $|b\rangle$ as an “identity measure”. The reliability can be achieved using a straightforward quantum process known as a swap test (Schuld, Sinayskiy & Petruccione, 2014). Liu & Rebentrost (2018) used kernel principal component analysis (PCA) and one-class support vector machine (SVM) quantum machine learning for quantum anomaly detection. They confirmed that using these algorithms is potentially appropriate to use in big quantum data applications (Liu & Rebentrost, 2018). By using two quantum classifiers, a variational quantum circuit and the quantum kernel estimator, Raubitzek & Mallinger (2023) investigated quantum machine learning classification tasks. They introduced a new dataset inspired by the concepts of quantum mechanics using the exponentially growing map of Linear algebra and demonstrated that both algorithms outperformed the classical machine learning algorithms (Raubitzek & Mallinger, 2023). A Quantum Neural Network for image classification is proposed by Senokosov et al. (2023) by utilising the medical MNIST and CIFAR 10 dataset and presented advancement in quantum machine learning by achieving accuracy 99 percent. Narottama, Mohamed & Aissa (2023) addressed challenges in deploying QML for wireless communications and proposed potential solutions to ensure feasibility and scalability. Chen et al. (2024) explored QML applications in image recognition, drug efficacy prediction, and cybersecurity, highlighting its potential advantages over classical machine learning and addressed key challenges in QML, including algorithm design, hardware limitations, data encoding, quantum landscapes, noise, and decoherence. Dai et al. (2024) proposed deep reinforcement learning (RL) to optimise QML architectures for supervised learning tasks, eliminating the need for predefined ansatz by an adaptive mechanism is integrated to dynamically adjust learning objectives, ensuring continuous improvement in the RL agent’s model discovery process. Extensive numerical simulations demonstrated that the proposed method effectively identifies Variational Quantum Circuit (VQC) architectures with high classification accuracy and minimal gate depth, advancing AI-driven quantum circuit design (Dai et al., 2024). Dutta et al. (2025) proposed a model that combined neural networks and

parameterised quantum circuits to predict tsunamis. Their proposed work got 96.01% accuracy, superior to the deep learning model (Dutta et al., 2025). Elsedimy, Elhadidy & Abohashish (2024) presented a novel intrusion detection model, QSVM-IGWO, to improve the detection capabilities and reduce false positive alarms of Hybrid Intrusion Detection System (HIDS). The Wolf Optimizer algorithm enhances the searching procedure and turns off Grey Wolf Optimizer (IGWO) limitations. The QSVM model is utilised for classification by selecting the kernel function to obtain an optimal solution. Experimental results show promising performance of QSVM-IGWO (Elsedimy, Elhadidy & Abohashish, 2024). Bilal et al. (2025) proposed a Q-BGWO-SQSVM algorithm that utilises an improved quantum-inspired binary Grey Wolf Optimizer and combines it with SqueezeNet and Support Vector Machines to exhibit sophisticated performance for breast cancer diagnosis. This approach received a remarkable performance of 99% accuracy and outperformed the state-of-the-art algorithm (Bilal et al., 2025). Liu et al. (2025), introduced quantum train framework for classification to revolutionise the machine learning by reducing the generalization errors showcasing the potential of quantum machine learning. Kwon et al. (2025) introduced the Variational Quantum Classifier (VQC) to restore and capitalise on the symmetrical properties of data distributions without relying on resampling techniques using quantum circuits optimised to mitigate the asymmetries inherent in imbalanced datasets. The proposed method demonstrates consistently superior performance across diverse datasets (Kwon et al., 2025). Maragkopoulos et al. (2025), proposed an algorithm for reducing the dimensionality of a given dataset by taking into account the specific quantum embedding that comes after, making quantum machine learning with VQCs more versatile and effective on high-dimensional datasets. They showed that the targeted dimensionality reduction method significantly enhanced VQC's performance (Maragkopoulos et al., 2025). Jura & Udrescu (2025) designed an instance of Grover's quantum search algorithm to accelerate the search for the optimal parameters of a Neural Network during the training process, a task traditionally performed using the backpropagation algorithm with the gradient descent method. Chen et al. (2025) proposed an enhanced natural parameterised quantum circuit, that achieved the maximum parameter capacity of quantum systems by preserving the local structure of the original dataset. For making the encoding circuits more experiment-friendly, two near-optimal circuits that achieve near-maximum capacity are provided. They show that the proposed method overwhelms other encoding methods for multiple datasets, demonstrating its potential for machine learning tasks (Chen et al., 2025). Deep reinforcement learning was used by Dai et al. (2024) to tackle the challenge of QML model construction by investigating effective QML model architectures specifically designed for particular supervised learning tasks. The approach entailed training a reinforcement learning agent to formulate policies that enable the exploration of quantum machine learning models without a predefined ansatz. A dynamic adaptive mechanism to modify the learning objectives also promotes ongoing enhancement in agent learning (Dai et al., 2024). The growing impact of machine learning on quantum technologies was underlined by Krenn et al. (2023), showcasing its role in quantum computing, communication, and simulation. They explored recent advancements in using artificial intelligence for quantum

measurements, device calibration, and experimental discovery while addressing future challenges and possibilities (Krenn et al., 2023). Jadhav, Rasool & Gyanchandani (2022) offered a brief assessment of contemporary techniques and algorithms in quantum machine learning and their applicability in addressing real-world problems (Jadhav, Rasool & Gyanchandani, 2022). To augment the capabilities of the QML, including expert systems, Various traditional Machine Learning (ML) methods and techniques can be transformed into quantum machine learning (QML) (Tychola, Kalampokas & Papakostas, 2023). Haboury et al. (2023) investigated how hybrid quantum-classical machine learning can optimise emergency evacuation planning during natural disasters. By integrating quantum neural networks with classical models, the study improves pathfinding accuracy in dynamically changing environments, enhancing pathfinding accuracy in dynamically changed settings (Haboury et al., 2023). The design of adders and multiplexers with fundamental gates is successfully combined by Ahmadpour et al. (2025), enhancing Artificial Intelligence in MIS and lowering energy consumption and latency. Melnikov et al. (2023) discussed the growing field of quantum machine learning and its impact on artificial intelligence and quantum computing. They highlight hybrid quantum-classical neural networks as a promising approach to improving model accuracy and efficiency while addressing challenges like error mitigation in quantum devices (Melnikov et al., 2023). Bhatia, Saggi & Kais (2023) discussed how quantum machine learning can enhance drug discovery by improving the prediction of molecules' absorption, distribution, metabolism, excretion, and toxicity (ADME-Tox) properties. By integrating quantum kernel methods with classical machine learning models, their framework demonstrates improved accuracy and efficiency in pharmaceutical research, potentially reducing costs and accelerating drug development (Bhatia, Saggi & Kais, 2023). Table 3 summarizes the literature review of the articles discussed in this section.

METHODOLOGY

This study developed and evaluated a hybrid quantum machine learning approach combining a Quantum Support Vector Machine (QSVM) with a Quantum Neural Network (QNN) to classify the breast cancer dataset, as shown in Fig. 2. The methodology begins with preprocessing the Dataset by encoding categorical labels, normalising feature values, and performing stratified sampling to ensure balanced class distribution. Quantum feature maps, including Z Feature Map, ZZ Feature Map, and Pauli Feature Map, transform classical data into quantum states suitable for kernel-based quantum classification. The Fidelity Quantum Kernel computes a quantum kernel matrix by measuring the overlap (fidelity) between quantum states, which is then used to train a classical support vector machine in a QSVM framework. The hybrid QSVM + QNN pipeline is simulated using Qiskit's Aer simulator to evaluate its robustness, accuracy, and learning capability. This hybrid framework is designed to leverage the geometric representation of QSVMs and the expressive capacity of QNNs to outperform individual quantum models on low-dimensional datasets under hardware-constrained quantum environments.

Here is the pseudo-code representation of the methodology:

```

BEGIN
LOAD dataset
APPLY preprocessing:
    - Handle imbalance (SMOTE or undersampling)
    - Standardise features
    - PCA → select top-k features
SPLIT data:
    train_set, external_test_set
QSVM PIPELINE
FOR each feature_map IN {Z, ZZ, Pauli} DO
    DEFINE quantum feature map (depth = d)
    PERFORM 5-fold CV:
        FOR each fold:
            embed fold data using feature map
            compute quantum kernel matrix
            train classical SVM on kernel
            record metrics
    TRAIN final QSVM on full train_set
    compute full training kernel
    fit SVM
EVALUATE on external_test_set:
    compute external kernel
    ext_acc, ext_prec, ext_rec, ext_f1, ext_auc
STORE:
    external_metrics, confusion_matrix, kernel_time
END FOR
QNN PIPELINE
FOR each feature_map IN {Z, ZZ, Pauli} DO
    DEFINE Quantum Neural Network:
        - embedding = feature_map
        - variational_layers = ansatz (L layers)
        - optimizer = Adam or SPSA
    PERFORM 5-fold CV:
        FOR each fold:
            train QNN classifier on train fold

```

(continued)

validate on val fold

record metrics

TRAIN final QNN on full train_set

measure train_time

EVALUATE on external_test_set:

ext_acc, ext_prec, ext_rec, ext_f1, confusion_matrix

STORE:

metrics, train_time

END FOR

HYBRID QSVM-QNN PIPELINE

FOR each feature_map IN {Z, PauliZ, ZZ}:

REPEAT experiment 3 times:

- Compute quantum kernel (QSVM part)*
- Extract quantum distances or support vectors*
- Pass kernel-derived features into QNN (hybrid part)*
- Train hybrid model on training set*
- Evaluate on external_test_set*

STORE:

acc_ext, prec_ext, rec_ext, f1_ext

confusion_matrix

kernel_time

t_test (SVM kernel time)

t_fq (QNN forward pass time)

END FOR

COMPARISON & ANALYSIS

END

Data preprocessing

We used the Breast Cancer Wisconsin dataset in this study available at <https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>.

Table 4 represents the full description of the dataset utilised and pair plot visualizations of some selected features of the dataset are shown in Fig. 3.

Let $X \in R^{n \times d}$ Be the input feature matrix, where n is the number of samples (selected for quantum feasibility), and d is the number of features used.

Rationale for dataset selection: Although larger datasets are often desirable for improving generalizability, the computational behaviour of quantum kernels places practical limits on scalability in simulator-based studies. In this experiment even a smaller dataset of 569 samples and seven features resulted in extremely high computation time.

Table 3 A literature review of the existing quantum assisted ML algorithms.

Ref.	Technique	Dataset	Accuracy	Platform used	Future work
<i>Ruskanda et al. (2023), Agarwal & Alam (2022)</i>	Quantum SVM	Dataset from UCI	98	Qiskit library	QSVM can be improved by using a different kernel function
<i>Jadhav, Rasool & Gyanchandani (2022), Bilal et al. (2025)</i>	Sqeeznet-SVM	Breast cancer	99	Matlab+GPU	Expand this research to a wider variety of datasets
<i>Abuga (2023), Elsedimy, Elhadidy & Abohashish (2024)</i>	Quantum SVM	Bot-IoT dataset	99.34	Intel i7-3540 processor	Further integration of ML and AI
<i>Tychola, Kalampokas & Papakostas (2023), Ajagekar & You (2021)</i>	QC-based hybrid deep learning model	Fault diagnosis dataset	N/A	Simulink tool in MATLAB	Fault diagnosis in larger systems
<i>Haboury et al. (2023), Batra et al. (2021)</i>	Quantum SVM	(SARS-CoV-2)	61	IBM's ibmq_rochester	Using other QML models
<i>Ahmadpour et al. (2025), Bhatia, Saggi & Kais (2023)</i>	Quantum SVM	ADME-Tox databases	74	qasm_simulator (qasm) statevector_simulator (SV)	N/A
<i>Wang (2023), Bishwas, Mani & Palade (2020)</i>	Quantum SVM	Breast cancer dataset	90	qasm_simulator	Availability of high number of qubits
<i>Fan, Shi & Zhu (2024), Singh et al. (2022)</i>	Quantum SVM	MNIST	97.5	qasm_simulator	Feature mapping using Quantum computer
<i>Li, Song & Wang (2021), Uehara et al. (2021)</i>	Quantum NN	NREL dataset	93.80	Qiskit SDK	Reducing quantum measurement error
<i>Ventura & Martinez (1998), Wang et al. (2019)</i>	Quantum KNN	Handwritten dataset	N/A	Quantum simulator	Applying classical neural network to QC
<i>Maragkopoulos et al. (2025), Xiao et al. (2010)</i>	Quantum+GA+K means clustering	3 simulated + 4 UCI datasets	N/A	Quantum simulator	Exploring search space using a small no. of solutions
<i>Li et al. (2022), Patel et al. (2019)</i>	Quantum NN	Original and fake signatures of a few people	95	Matlab+java	Online signature verification
<i>Melnikov et al. (2023), Sharma (2020)</i>	Quantum KNN	Breast cancer dataset	97	Qiskit SDK	Improving efficiency and scalability of quantum machine learning algorithms
<i>Jura & Udrescu (2025), Sharma, Singh & Kumar (2025)</i>	Quantum decision tree	Wisconsin breast cancer, Wheat seed dataset, Haberman's cancer survival	90, 90, 100	Quantum simulator	N/A
<i>Akrom (2024), Masum et al. (2023)</i>	Quantum SVM	The Bengali dataset, Twitter datasets	72, 84	qasm simulator	Exploring error mitigation and noise correction techniques
<i>Bhatia, Saggi & Kais (2023), Mercaldo et al. (2022)</i>	Quantum NN	Android malware detection dataset	95	Tensorflow Quantum	Improvement of the proposed method

For example, Z feature map required (400–800 s per fold), while ZZ and Pauli feature maps are required (4,500–8,400 s per fold). This occurs because QSVM requires $N \times N$ kernel matrix calculations. Where computational cost grows quadratically ($O(N^2)$) with respect

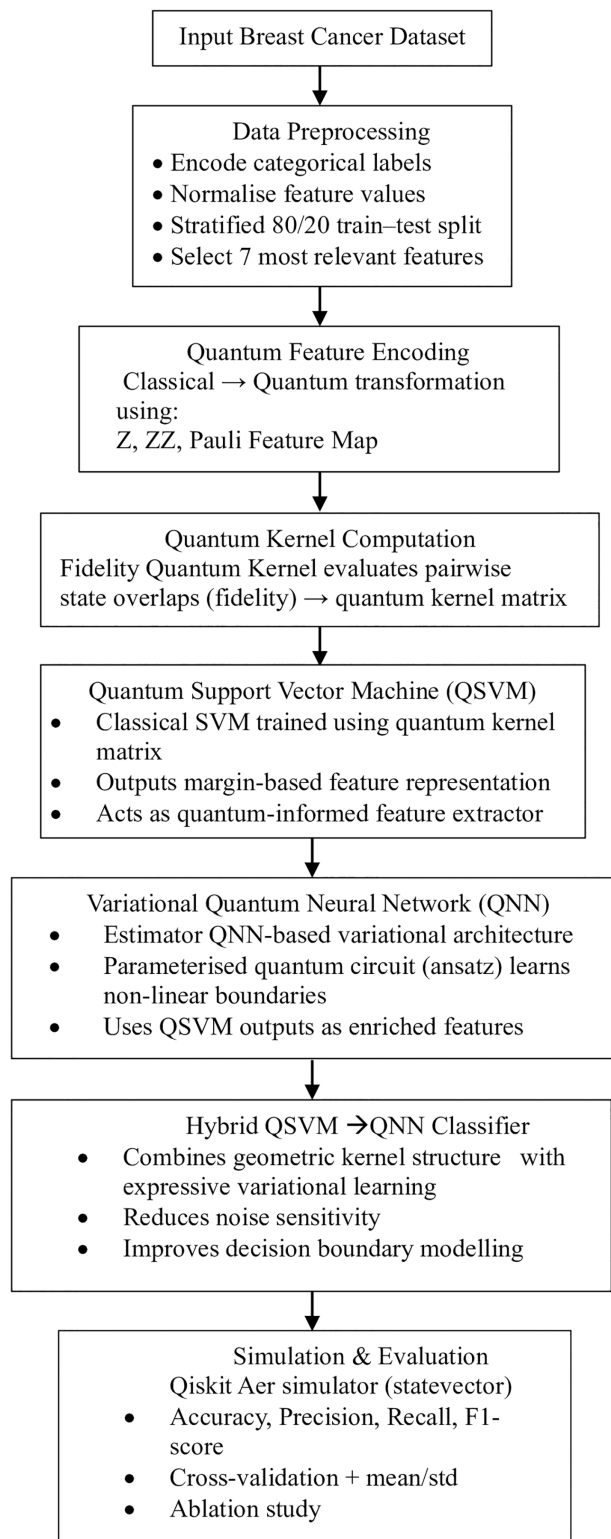


Figure 2 The proposed methodology.

Full-size DOI: 10.7717/peerj-cs.3600/fig-2

Table 4 Full description of dataset.

Characteristic	Detail
Number of instances	569 (357 Benign, 212 Malignant)
Number of attributes	30 numeric attributes and 1 class label
Classes	Benign (B), Malignant (M)
Missing attribute values	None
Class distribution	62.7% Benign, 37.3% Malignant
Creator	Dr. William H. Wolberg
Donor	University of Wisconsin-Madison
Date	1995

to the number of samples. Any increase in dataset size causes the quantum kernel to expand drastically, pushing execution time in hours and even in days.

Given that the current experiments already taking thousands of seconds, so this dataset is chosen to balance scientific validity with computational feasibility.

Label encoding

This technique converts the categorical target labels into numerical format, which machine learning models and quantum classifiers require.

If we are using the breast cancer dataset, then label encoding

–*Label Mapping*: $\{Benign \rightarrow 0, Malignant \rightarrow 1\}$

Let, $y \in \{Benign, Malignant\}$ Then, the encoded label is

$$y_{encoded} \in \{0, 1\}$$

Feature scaling

Quantum models are susceptible to the scale of input features. To prevent feature dominance and ensure numerical stability during circuit embedding (*via* feature maps), Z-score normalisation is applied to rescale each feature x_i With zero mean and unit variance.

$$X_{scaled} = \frac{X - \mu}{\sigma}$$

where $X \in R^{n \times d}$ is the original dataset with n samples and d Features. $\mu \in R^d$ is the mean vector computed across the training data for each feature. $\sigma \in R^d$ is the standard deviation vector for each feature ([Ajagekar & You, 2021](#); [Benedetti et al., 2019](#)).

Quantum SVM

QSVM utilises quantum kernel methods for classification by mapping classical data into a higher-dimensional Hilbert space using quantum feature maps. The classification uses a classical Support Vector Machine (SVM) with a precomputed quantum kernel matrix.

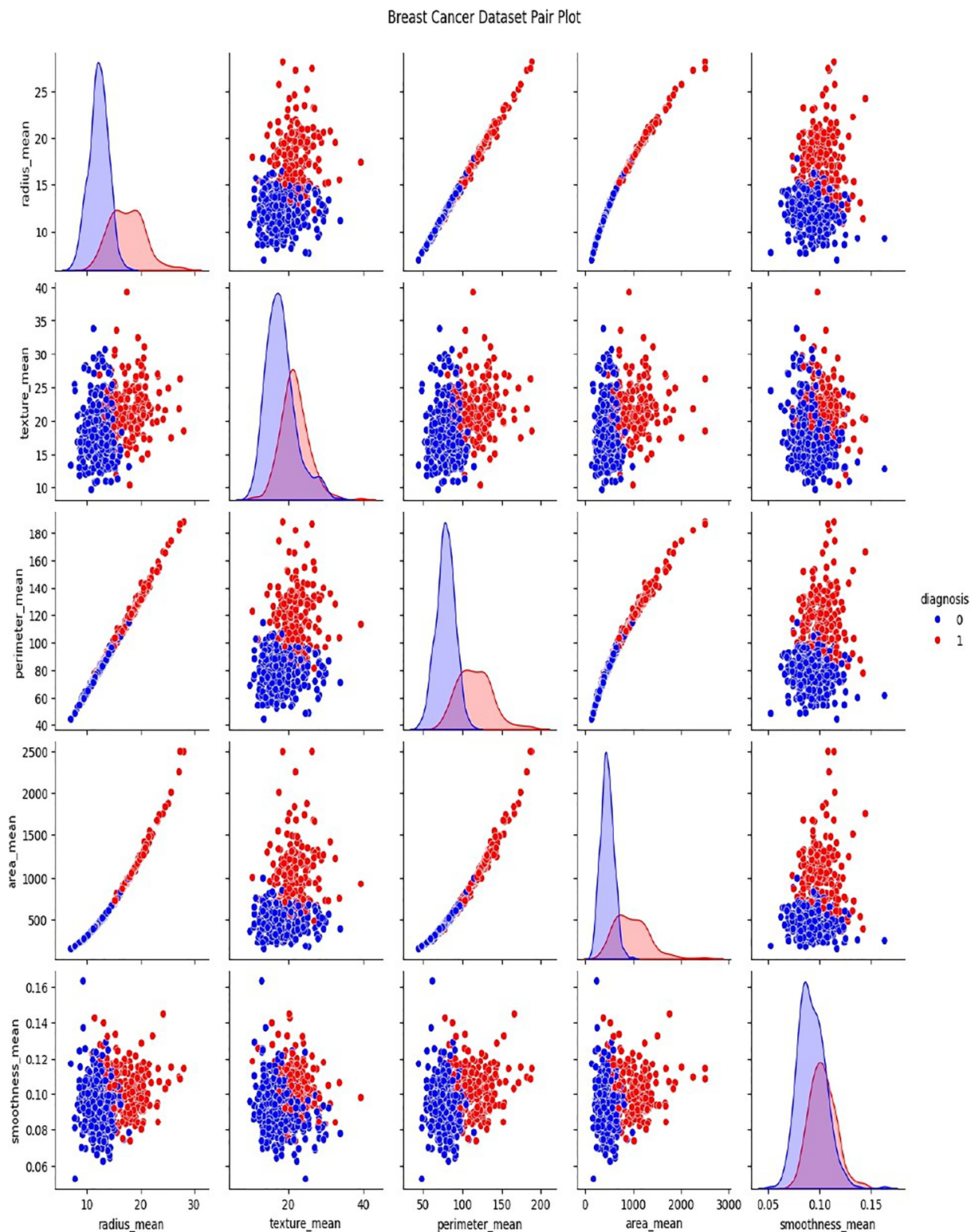


Figure 3 Pair plot visualizations of some selected features of dataset.

Full-size DOI: 10.7717/peerj-cs.3600/fig-3

Global Phase: $3.0 \cdot x[0] + 3.0 \cdot x[1] + 3.0 \cdot x[2] + 3.0 \cdot x[3] + 3.0 \cdot x[4] + 3.0 \cdot x[5] + 3.0 \cdot x[6]$

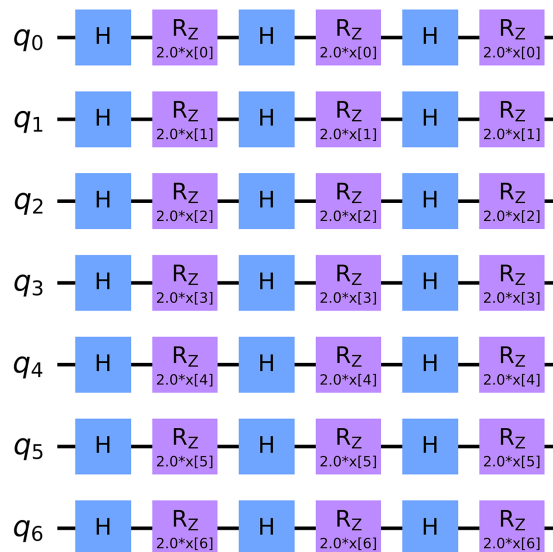


Figure 4 Z feature map circuit.

Full-size DOI: 10.7717/peerj-cs.3600/fig-4

Feature encoding

A quantum feature map is a unitary transformation $U_{\phi(x)}$ That encodes classical input vectors into quantum states. The used feature map in this study is Z, ZZ and Pauli Feature Map, which are explained as follows:

Z Feature Map (phase encoding along z axis): For a given input vector $x = x_1, x_2, \dots, x_n$ where each $x_i \in \mathbb{R}$ then z feature map of x_i is shown in Fig. 4:

$$RZ(\phi_i) = \exp(-ix_i \frac{\sigma_z}{2}). \quad (1)$$

ZZ Feature map (entangled phase encoding): for every pair of qubits (i, j) it applies $RZ_Z(x_i \cdot x_j)$.

$$RZ_Z(x_i \cdot x_j) = \exp(-ix_i x_j Z \otimes Z / 2) \quad (2)$$

is shown in Fig. 5.

Pauli Feature map (multi axis rotational map): It chooses the combinations of Pauli strings (shown in Fig. 6) P where $P \in \{X, Y, Z\}^{\otimes n}$ (Batra et al., 2021; Schuld et al., 2020)

$$U_{(x)} = \exp(-i\phi(x) \cdot P). \quad (3)$$

Fidelity kernel matrix computation

The quantum kernel function is computed by evaluating the fidelity between quantum states:

$$K(x, x') = \langle \psi(\frac{\rightarrow}{x}) | \psi(\frac{\rightarrow}{x'}) \rangle^2. \quad (4)$$

Global Phase: $3.0*x[0] + 3.0*x[1] + 3.0*x[2] + 3.0*x[3] + 3.0*x[4] + 3.0*x[5] + 3.0*x[6] + 3.0*(\pi - x[0])*(\pi - x[1]) + 3.0*(\pi - x[0])*(\pi - x[2]) + 3.0*(\pi - x[0])*(\pi - x[3]) + 3.0$

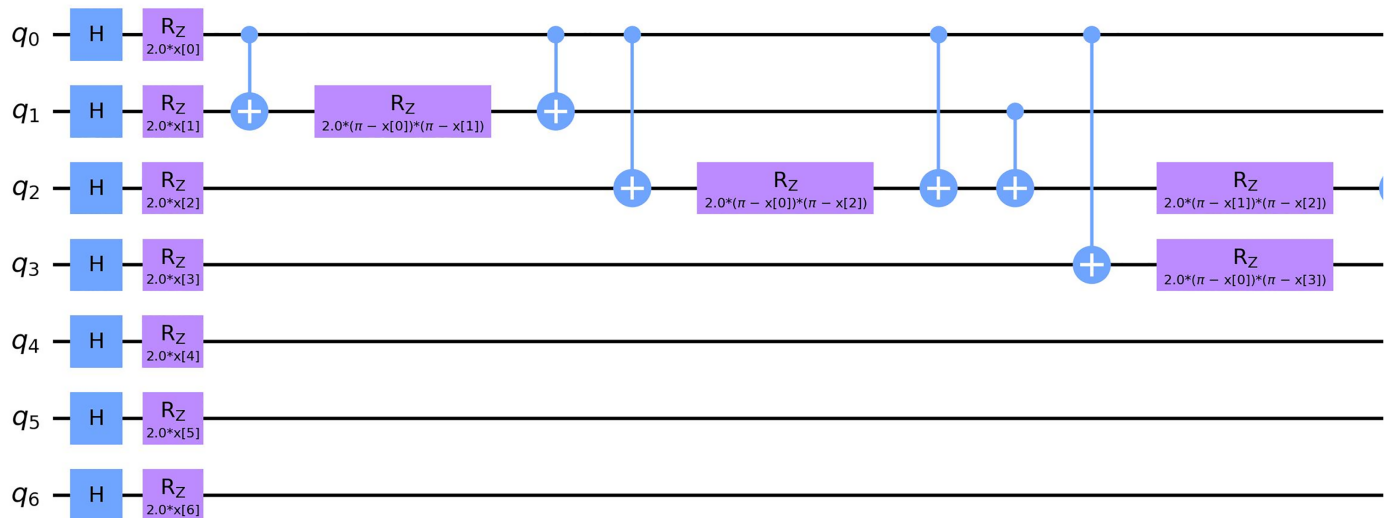


Figure 5 ZZ feature map circuit.

Full-size DOI: 10.7717/peerj-cs.3600/fig-5

Global Phase: $3.0*x[0] + 3.0*x[1] + 3.0*x[2] + 3.0*x[3] + 3.0*x[4] + 3.0*x[5] + 3.0*x[6]$

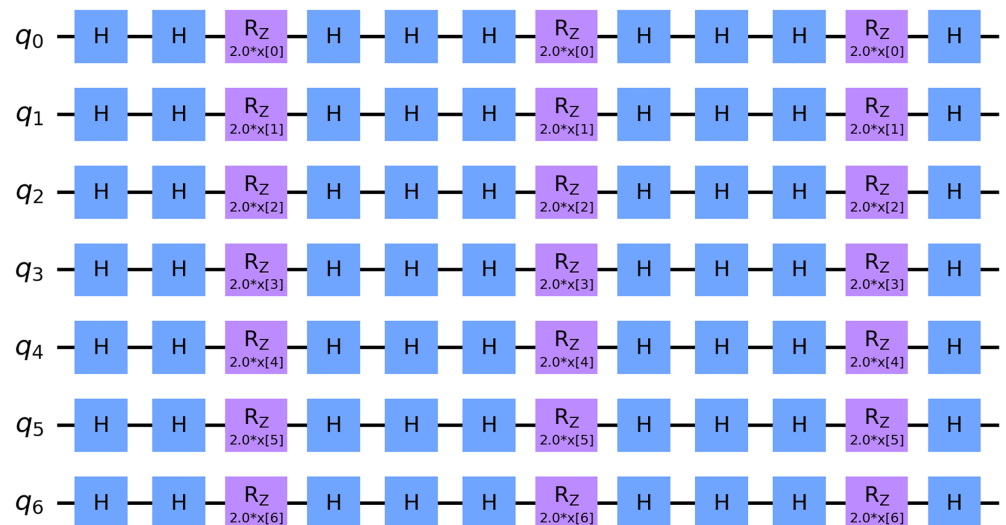


Figure 6 Pauli feature map circuit.

Full-size DOI: 10.7717/peerj-cs.3600/fig-6

This kernel matrix is used to train the classical SVM model. The training and testing kernel matrices are defined as:

$$K_{train} \in R^{m \times m}, \quad K_{test} \in R^{m \times m} \quad (5)$$

where m is the number of training samples and k is the number of test samples.

The kernel is implemented using the Fidelity Quantum Kernel from Qiskit, which evaluates the overlaps between quantum states.

SVM classifier

Using the precomputed quantum kernel matrix, we train a Support Vector Machine (SVM) with a precomputed kernel function:

$$f(x) = \text{sign} \left(\sum_{i=1}^m \alpha_i y_i K(x_i, x) + b \right) \quad (6)$$

where α_i are the Lagrange multipliers obtained from the SVM optimisation, y_i are the class labels, $K(x_i, x)$ is the quantum kernel value between training and test samples and b is the bias term.

The Quantum Support Vector Machine algorithm classifies data by evaluating the quantum kernel similarity between the test and training samples, rendering it suitable for datasets with complex feature interrelations (Bishwas, Mani & Palade, 2020; Gentinetta et al., 2024).

Rationale for hybrid QSVM and QNN

QSVMs and QNNs each exhibit complementary strengths and weaknesses when applied independently to real-world datasets on near-term quantum hardware. The initial experiments outlined that QSVM consistently achieved strong separability due to their kernel-based structure and expressive quantum feature maps; but they also incurred very high computational cost, particularly during the full kernel matrix evaluation $O(N^2)$, which scales quadratically with the number of samples. In contrast, the QNN demonstrated significantly lower execution time, but their standalone performance on the Breast Cancer dataset remained unstable and accuracy-limited, primarily due to the small number of trainable parameters available in shallow two-qubit ansatz.

The motivation behind the development of the hybrid approach using QSVM and QNN is the findings from the initial experiments of standalone QML algorithms. This approach combines the benefits of both models while compensating for their individual limitations. The QSVM is strictly used as a quantum feature extractor, generating high margin data information and capturing complex geometry in the data through quantum feature maps. This step maximizes the expressive capacity of the model while limiting QSVM to a single forward computation of the quantum kernel, rather than repeatedly performing inference during classification.

The output of the QSVM margin is then combined with low dimensional classical embeddings using PCA and fed into a compact two qubit QNN. Since the dimensionality of input is drastically reduced by the QSVM, the QNN can be efficiently trained and experience no vanishing gradient issues which are associated with high dimensional encodings and require only a small number of steps overall. The QNN works effectively as a lightweight classifier by sharpening the QSVM's coarse decision boundary and providing final class prediction.

By assigning each computational task according to the capabilities of each model-QSVM for high quality feature extraction and QNN for efficient classification task-we achieved a hybrid pipeline that significantly reduces the overall execution time

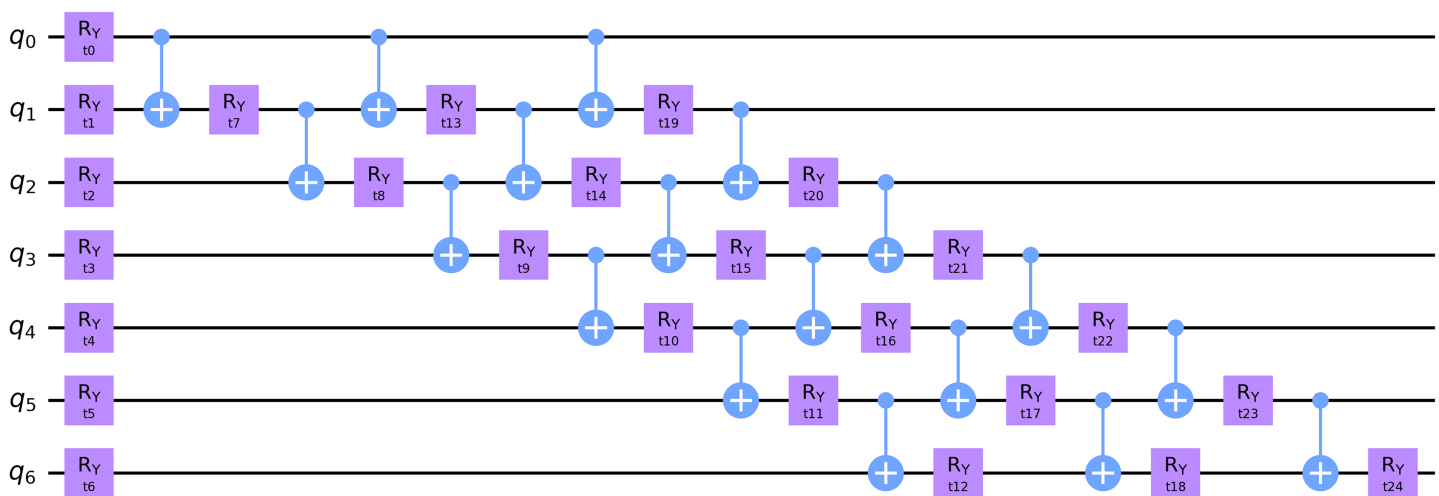


Figure 7 Quantum circuit of hybrid QNN approach.

Full-size DOI: 10.7717/peerj-cs.3600/fig-7

while improving classification accuracy simultaneously. The circuit diagram in Fig. 7 illustrates seven features of breast cancer dataset using seven qubits.

Quantum Kernel feature extraction using QSVM

The first stage of our pipeline involves transforming the input data x_i Using a quantum kernel function computed from a quantum feature map:

$$z_i = f(K(x_i, X_{train})) \quad (7)$$

where x_i is the input data sample. $(K(x_i, X_{train}))$ is the Fidelity Quantum Kernel, computed as:

$$K(x, x') = \langle \psi(\vec{x}) | \psi(\vec{x}') \rangle^2 \quad (8)$$

z_i represents the QSVM-transformed feature vector, capturing quantum-enhanced feature relationships (Sharma, 2020; Yin, 2024).

After the kernel is computed, a SVM is trained using the transformed data. However, instead of using the SVM classifier for final predictions, the intermediate kernel features to a QNN are passed for improved decision boundaries.

Classification using quantum neural network (QNN)

After feature extraction, the transformed features z_i are fed into an EstimatorQNN that learns a quantum-parametrized function to classify the data. The QNN output is computed as:

$$\hat{y}_i = \arg \max p(y|z_i, \theta) \quad (9)$$

where $\hat{y}_i$ is the predicted label, $p(y|z_i, \theta)$ is the probability distribution over class labels. θ represents trainable variational parameters of the quantum circuit.

Using QSVM as a feature extractor, we retain the advantages of quantum kernels, while the QNN refines classification boundaries that QSVM alone might struggle with. This

combination is beneficial for high-dimensional datasets where kernel-based methods alone may not provide optimal decision boundaries.

Simulator

For ensuring reproducibility and high-fidelity benchmarking of quantum machine learning (QML) models, all experiments were conducted on Qiskit Aer simulator, which emulate quantum hardware behaviour on classical computing infrastructure. Specifically, we used the following simulation backends:

State vector simulator

It is beneficial for evaluating the FidelityQuantumKernel, where the inner product between quantum states is essential for kernel matrix computation in QSVM. In this study:

Given two quantum states $|\psi(\vec{x})\rangle$ and $|\psi(\vec{x}')\rangle$ The fidelity kernel is defined as:

$$K(x, x') = \langle \psi(\vec{x}) | \psi(\vec{x}') \rangle^2. \quad (10)$$

This quantity is computed directly from the state vector representation of each input sample.

Estimator QNN

The Estimator Quantum Neural Network (Estimator QNN) is a key component in our hybrid QSVM-QNN classification framework. After the QSVM extracts kernel-transformed features from the dataset, these features are passed to the Estimator QNN, which learns an optimal decision boundary for classification.

The Estimator QNN is built using a parameterised quantum circuit $U(\theta)$ With trainable variational parameters:

$$|\psi\theta\rangle = U(z_i, \theta)|0\rangle. \quad (11)$$

The output prediction is computed as the expectation value of a measurement operator M .

$$f_\theta(z_i) = \langle \psi_\theta | M | \psi_\theta \rangle. \quad (12)$$

Evaluation metrics

The following standard classification metrics are employed:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (13)$$

$$Precision = \frac{TP}{TP + FP} \quad (14)$$

$$Recall = \frac{TP}{TP + FN} \quad (15)$$

$$F1\text{-score} = \frac{2 \cdot precision \cdot recall}{precision + recall} \quad (16)$$

$$\text{Confusion matrix} = \begin{bmatrix} TP & FN \\ FP & TN \end{bmatrix}. \quad (17)$$

Evaluation method

The evaluation of the proposed Quantum Machine Learning Model was conducted through a systematic and rigorous procedure to ensure reliability and reproducibility. The following quantum methods were employed:

Baseline comparison with classical model

To establish a performance benchmark, the proposed model is directly compared with machine learning models and quantum machine learning algorithms. The comparative analysis served as a baseline validation method to assess the relative advantage of quantum models.

Feature map ablation study

An ablation study was conducted by systematically varying the quantum feature maps, namely Z, ZZ and Pauli. This procedure enables the isolation of the contribution of the Feature mapping scheme to the overall performance.

Computational time comparison across quantum feature maps

The computational cost of quantum feature maps is compared in this procedure.

Computing infrastructure

All the simulations were performed using IBM's local simulator, namely the state vector simulator, on a dedicated personal computing environment. The local system was configured with the Windows 10 operating system, an Intel i5 processor and 16 GB of RAM. This setup ensures consistent performance for running quantum simulations and facilitates reproducibility of results.

RESULTS AND DISCUSSION

We evaluated the performance of the proposed hybrid quantum classifier, which combines a QSVM as a feature extractor and a QNN as the final classifier on the Breast Cancer Wisconsin Diagnostic dataset. Three quantum feature maps, Z Feature Map, ZZ Feature Map and Pauli Feature Map, were tested, transforming the classical data into a quantum Hilbert space *via* the Fidelity Quantum Kernel. Seven features were extracted using PCA. The seven features of the breast cancer dataset extracted are (mean concavity, mean perimeter, worst concave points, mean concave points, worst radius, worst area, worst perimeter). All experiments followed a consistent pipeline that included z score normalization, an 80–20% stratified train test split and repeated 5-fold cross validation. The QSVM solely operated as a feature extractor, producing decision function margins that were subsequently compressed *via* PCA to a 2-dimensional embedding suitable for a 2-qubit QNN with a real amplitude ansatz. Performance was assessed using accuracy, precision, recall, F1-score and confusion matrices, with separate reporting for (1) internal cross validation and (2) external testing on the unseen 20% hold out set.

Table 5 Proposed model parameters and hyperparameters.

Parameter	Values
Dataset	Breast Cancer Wisconsin diagnostic and wine dataset (sklearn version)
Target variable	Diagnosis (Benign = 0, Malignant = 1)
Feature selection	Mutual information (Top-7 highest scoring features)
Normalization	StandardScaler (Z-score) – fitted on training data only
Train–Test split	80% train/20% test, stratified, random_state = 42
Kernel dimensionality	7-dimensional QSVM kernel (no PCA on kernel stage)
Embedding for QNN	Margin (SVM decision function) + PCA (1 component) → 2-dimensional input for QNN
Cross-validation	5-fold StratifiedKFold, shuffle = True, random_state = seed
No. of repetitions	3 independent repeats (seed = 42, 43, 44)
Quantum simulators	AerSimulator (method = “statevector”)
Quantum feature maps	ZFeatureMap, PauliFeatureMap (Z-only), ZZFeatureMap
Feature map settings	feature_dimension = 7, reps = 1, full entanglement
Quantum kernel	Fidelity Quantum Kernel: QuantumKernel (feature_map=...)
QSVM classifier	SVC (kernel = ‘precomputed’, C = 1)
QSVM role	Feature extractor only (decision margins used; NOT used for final prediction)
QNN Ansatz	RealAmplitudes (num_qubits = 2, reps = 1, entanglement = “full”)
QNN model	EstimatorQNN + NeuralNetworkClassifier
QNN optimizer	COBYLA (maxiter = 80)
Timing metrics recorded	Kernel computation time, SVM training time, QNN training time
External evaluation	Hold-out test set (20%) evaluated after each repeat
Output metrics	Accuracy, Precision, Recall, F1-score, Confusion matrix
Total experimental runs	3 repetitions × 3 feature maps × 5 CV folds = 45 QSVM–QNN models + 3 external evaluations

Table 6 List of parameters of classical machine learning classification.

Model	Hyperparameters (Grid Search Range)
SVM	$C \in \{0.1, 1, 10\}$; Kernel $\in \{\text{rbf}, \text{linear}\}$
Random forest	n_estimators $\in \{100, 200\}$; max_depth $\in \{\text{None}, 5, 10\}$
KNN	n_neighbors $\in \{3, 5, 7\}$
Decision tree	max_depth $\in \{\text{None}, 5, 10\}$
Logistic regression	$C \in \{0.1, 1, 10\}$; max_iter = 2,000
Common preprocessing	StandardScaler() applied to all models
Validation strategy	Nested CV: 5 outer folds, 3 inner folds

Experimental configuration

Tables 5 and 6 summarise the key parameters and configurations used in the classification model applied to the breast cancer dataset. The selected features, quantum kernel setting, classifier and evaluation metrics are outlined, providing a clear overview of the experimental setup. A total of 45 QSVM→QNN models were trained across three feature maps, three repetitions and five folds per repetition.

Standalone QSVM performance

Tables 7 and 8 report the cross validation and external test performance for the individual QSVM model. The Z feature map consistently achieved the strongest performance, with cross validation accuracy (0.89–0.94), external accuracy (0.947) and confusion matrix [[38, 4], [2, 70]].

In contrast the ZZ and Pauli feature map produced noticeable lower external accuracies around (0.86), accompanied by substantial increase in kernel computational time due to kernel evaluation with entangling feature maps required 10–15 times longer than Z feature map.

Major cons of QSVM:

- Poor NISQ scalability: kernel matrices scale as $O(N^2)$ with dataset size $\rightarrow$ completely impractical for high dimensional or large sample problems. Noise accumulates heavily when computing kernel overlaps.
- Extremely high kernel computational cost: kernel time increases with 10–15 times. This makes QSVM unsuitable for larger datasets, deeper circuits, real time or resources limited applications.

Standalone QNN performance

Tables 9 and 10 presents the QNN results. In contrast to QSVM, the QNN showed weaker generalisations ability with external accuracy (0.49–0.60) (depending upon the feature maps), highly unstable precision and recall and training time (120–150 s) for Z feature map and almost 1,000 s for entangling maps. The QNN frequently overfitted the training folds, resulting in a substantial drop in performance on the external test set. The entangling feature maps further degraded performance owing to gradient vanishing and optimization instability.

Major cons of QNN:

- Poor generalizations on external data, QNN overfits the training data due to the variational parameters.
- High sensitivity to noise and barren plateaus, Z performed reasonably, but Pauli and ZZ collapses in performance due to deeper entanglement, vanishing gradient and parameter noise amplification.
- Optimization instability: Training landscape is highly non convex and fragile. Different runs vary widely before convergence.

Hybrid QSVM $\rightarrow$ QNN results

Tables 11 and 12 leverages the QSVM solely for quantum feature extraction, using the decision-function margin as a 1-dimensional nonlinear embedding. This margin is further combined with PCA (one component), yielding a 2-dimensional input to the QNN.

Across all repeats and feature maps, the hybrid model showed improved stability relative to stand-alone QNNs and partially retained the discriminative power of the QSVM

Table 7 Results of QSVM using breast cancer dataset (cross-validation accuracy).

Map	Repeat	Fold	Accuracy	Precision	Recall	F1-score	Kernel time
Z	0	1	0.912088	0.929825	0.929825	0.929825	419.6922
Z	0	2	0.945055	0.919355	1	0.957983	479.7656
Z	0	3	0.89011	0.898305	0.929825	0.913793	427.8788
Z	0	4	0.89011	0.912281	0.912281	0.912281	430.5173
Z	0	5	0.923077	0.916667	0.964912	0.940171	420.3765
Z	1	1	0.901099	0.887097	0.964912	0.92437	398.2531
Z	1	2	0.901099	0.928571	0.912281	0.920354	369.323
Z	1	3	0.901099	0.875	0.982456	0.92562	369.8072
Z	1	4	0.945055	0.919355	1	0.957983	369.4248
Z	1	5	0.89011	0.912281	0.912281	0.912281	384.598
Z	2	1	0.901099	0.887097	0.964912	0.92437	380.0891
Z	2	2	0.867033	0.865517	0.882456	0.873913	397.097
Z	2	3	0.923077	0.903226	0.982456	0.941176	393.8337
Z	2	4	0.868132	0.894737	0.894737	0.894737	388.8237
Z	2	5	0.912088	0.945455	0.912281	0.928571	394.4727
ZZ	0	1	0.89011	0.898305	0.929825	0.913793	4,525.521
ZZ	0	2	0.868132	0.857143	0.947368	0.9	4,505.814
ZZ	0	3	0.868132	0.868852	0.929825	0.898305	4,502.868
ZZ	0	4	0.89011	0.898305	0.929825	0.913793	4,497.659
ZZ	0	5	0.846154	0.84127	0.929825	0.883333	4,492.259
ZZ	1	1	0.934066	0.932203	0.964912	0.948276	4,501.059
ZZ	1	2	0.857143	0.833333	0.964912	0.894309	4,496.399
ZZ	1	3	0.857143	0.866667	0.912281	0.888889	4,503.443
ZZ	1	4	0.901099	0.863636	1	0.926829	4,490.462
ZZ	1	5	0.857143	0.866667	0.912281	0.888889	4,632.36
ZZ	2	1	0.912088	0.888889	0.982456	0.933333	4,516.301
ZZ	2	2	0.857143	0.866667	0.912281	0.888889	4,500.656
ZZ	2	3	0.879121	0.859375	0.964912	0.909091	4,485.379
ZZ	2	4	0.857143	0.866667	0.912281	0.888889	4,607.22
ZZ	2	5	0.89011	0.861538	0.982456	0.918033	4,510.006
Pauli	0	1	0.89011	0.898305	0.929825	0.913793	4,525.015
Pauli	0	2	0.868132	0.857143	0.947368	0.9	4,495.28
Pauli	0	3	0.868132	0.868852	0.929825	0.898305	4,498.907
Pauli	0	4	0.89011	0.898305	0.929825	0.913793	4,501.673
Pauli	0	5	0.846154	0.84127	0.929825	0.883333	4,495.358
Pauli	1	1	0.934066	0.932203	0.964912	0.948276	4,490.307
Pauli	1	2	0.857143	0.833333	0.964912	0.894309	4,492.211
Pauli	1	3	0.857143	0.866667	0.912281	0.888889	4,491.087
Pauli	1	4	0.901099	0.863636	1	0.926829	4,491.443
Pauli	1	5	0.857143	0.866667	0.912281	0.888889	4,498.298
Pauli	2	1	0.912088	0.888889	0.982456	0.933333	4,508.257

Table 7 (continued)

Map	Repeat	Fold	Accuracy	Precision	Recall	F1-score	Kernel time
Pauli	2	2	0.857143	0.866667	0.912281	0.888889	8,487.27
Pauli	2	3	0.879121	0.859375	0.964912	0.909091	4,633.449
Pauli	2	4	0.857143	0.866667	0.912281	0.888889	4,548.08
Pauli	2	5	0.89011	0.861538	0.982456	0.918033	4,549.62

Table 8 Results of QSVM using breast cancer dataset (External accuracy).

Feature_Map	Ext_Acc	Ext_Prec	Ext_Rec	Ext_F1	Ext_Auc	Kernel_Time	Confusion_Matrix
Z	0.947368	0.945946	0.972222	0.958904	0.977183	616.7011	[[38, 4], [2, 70]]
ZZ	0.859649	0.858974	0.930556	0.893333	0.941799	7,255.629	[[31, 11], [5, 67]]
Pauli	0.859649	0.858974	0.930556	0.893333	0.941799	7,136.131	[[31, 11], [5, 67]]

Table 9 Results of QNN using breast cancer dataset (cross-validation accuracy).

Map	Repeat	Fold	Accuracy	Precision	Recall	F1-score	Kernel time
Z	0	1	0.6593	0.6970	0.8070	0.7480	118.57
Z	0	2	0.5714	0.6364	0.7368	0.6829	118.86
Z	0	3	0.6044	0.6364	0.8596	0.7313	118.20
Z	0	4	0.6044	0.6438	0.8246	0.7231	144.71
Z	0	5	0.6264	0.6667	0.8070	0.7302	142.86
Z	1	1	0.5714	0.6184	0.8246	0.7068	142.03
Z	1	2	0.6593	0.6806	0.8596	0.7597	149.44
Z	1	3	0.5824	0.6173	0.8772	0.7246	126.49
Z	1	4	0.6374	0.6765	0.8070	0.7360	127.12
Z	1	5	0.6154	0.6528	0.8246	0.7287	120.73
Z	2	1	0.6484	0.6842	0.8246	0.7480	132.41
Z	2	2	0.5934	0.6420	0.7895	0.7090	139.22
Z	2	3	0.6044	0.6571	0.8246	0.7304	141.73
Z	2	4	0.6264	0.6718	0.8070	0.7317	137.57
Z	2	5	0.6484	0.6956	0.7895	0.7399	133.48
ZZ	0	1	0.6044	0.6296	0.8947	0.7391	991.05
ZZ	0	2	0.6044	0.6400	0.8421	0.7273	986.29
ZZ	0	3	0.5495	0.6143	0.7544	0.6772	976.00
ZZ	0	4	0.5385	0.6027	0.7719	0.6769	969.39
ZZ	0	5	0.5604	0.6232	0.7544	0.6825	969.02
ZZ	1	1	0.5495	0.6053	0.8070	0.6917	983.37
ZZ	1	2	0.6044	0.6296	0.8947	0.7391	974.81
ZZ	1	3	0.5714	0.6216	0.8070	0.7023	969.40
ZZ	1	4	0.5714	0.6250	0.7895	0.6977	980.54
ZZ	1	5	0.5495	0.6111	0.7719	0.6822	980.99
ZZ	2	1	0.5824	0.6341	0.8246	0.7163	989.62

(Continued)

Table 9 (continued)

Map	Repeat	Fold	Accuracy	Precision	Recall	F1-score	Kernel time
ZZ	2	2	0.5934	0.6420	0.7895	0.7090	974.33
ZZ	2	3	0.5495	0.6078	0.7719	0.6814	969.77
ZZ	2	4	0.5604	0.6226	0.7719	0.6891	982.55
ZZ	2	5	0.5824	0.6403	0.7895	0.7071	978.13
Pauli	0	1	0.5824	0.6203	0.8596	0.7206	971.05
Pauli	0	2	0.5824	0.6338	0.7895	0.7031	970.36
Pauli	0	3	0.5714	0.6184	0.8246	0.7068	973.66
Pauli	0	4	0.5055	0.5769	0.7895	0.6667	972.77
Pauli	0	5	0.5824	0.6507	0.7193	0.6833	979.02
Pauli	1	1	0.5275	0.5946	0.7719	0.6718	994.95
Pauli	1	2	0.5495	0.6025	0.8246	0.6963	971.56
Pauli	1	3	0.5714	0.6324	0.7544	0.6880	968.76
Pauli	1	4	0.5714	0.6154	0.8421	0.7111	969.43
Pauli	1	5	0.5934	0.6316	0.8421	0.7218	972.08
Pauli	2	1	0.5604	0.6122	0.8246	0.7023	986.17
Pauli	2	2	0.5495	0.6049	0.7895	0.6842	971.44
Pauli	2	3	0.5385	0.5970	0.7544	0.6667	974.12
Pauli	2	4	0.5714	0.6274	0.7895	0.6997	979.29
Pauli	2	5	0.5824	0.6439	0.7895	0.7092	985.01

Table 10 Results of QNN using breast cancer dataset (external accuracy).

Feature_Map	Ext_Acc	Ext_Prec	Ext_Rec	Ext_F1	Train_Time	Confusion_Matrix
Z	0.605263	0.633663	0.888889	0.739884	148.7304	[[5, 37], [8, 64]]
ZZ	0.491228	0.606061	0.555556	0.57971	1,241.746	[[16, 26], [32, 40]]
Pauli	0.54386	0.606383	0.791667	0.686747	1,212.401	[[5, 37], [15, 57]]

Table 11 Results of hybrid QSVM+QNN using breast cancer dataset (cross-validation accuracy).

Map	Repeat	Fold	Acc	Prec	Rec	F1-score	Svm_Time	Qnn_Time	Kernel_Time
Z	1	1	0.89011	0.885246	0.947368	0.915254	0.004718	21.57321	708.9336
Z	1	2	0.945055	0.964286	0.947368	0.955752	0.005002	16.57699	708.9336
Z	1	3	0.901099	0.913793	0.929825	0.921739	0.004001	12.91412	708.9336
Z	1	4	0.945055	0.933333	0.982456	0.957265	0.003999	11.67949	708.9336
Z	1	5	0.934066	0.918033	0.982456	0.949153	0.003998	27.64144	708.9336
Z	2	1	0.923077	0.916667	0.964912	0.940171	0.032484	15.20942	420.9515
Z	2	2	0.824176	0.847458	0.877193	0.862069	0	21.83338	420.9515
Z	2	3	0.934066	0.918033	0.982456	0.949153	0.007702	19.79435	420.9515
Z	2	4	0.956044	0.949153	0.982456	0.965517	0	14.06794	420.9515
Z	2	5	0.901099	0.9	0.947368	0.923077	0.005395	15.68666	420.9515
Z	3	1	0.923077	0.931034	0.947368	0.93913	0.015849	68.11751	1,385.417
Z	3	2	0.923077	0.931034	0.947368	0.93913	0.007806	46.53129	1,385.417

Table 11 (continued)

Map	Repeat	Fold	Acc	Prec	Rec	F1-score	Svm_Time	Qnn_Time	Kernel_Time
Z	3	3	0.901099	0.928571	0.912281	0.920354	0	56.36003	1,385.417
Z	3	4	0.923077	0.890625	1	0.942149	0	53.32256	1,385.417
Z	3	5	0.934066	0.932203	0.964912	0.948276	0.0077	46.59176	1,385.417
PauliZ	1	1	0.89011	0.885246	0.947368	0.915254	0	28.22314	491.8407
PauliZ	1	2	0.945055	0.964286	0.947368	0.955752	0	21.96358	491.8407
PauliZ	1	3	0.901099	0.913793	0.929825	0.921739	0	15.67514	491.8407
PauliZ	1	4	0.945055	0.933333	0.982456	0.957265	0	14.53816	491.8407
PauliZ	1	5	0.934066	0.918033	0.982456	0.949153	0	32.51618	491.8407
PauliZ	2	1	0.923077	0.916667	0.964912	0.940171	0.007315	21.97719	491.0782
PauliZ	2	2	0.824176	0.847458	0.877193	0.862069	0	36.91044	491.0782
PauliZ	2	3	0.934066	0.918033	0.982456	0.949153	0	25.89656	491.0782
PauliZ	2	4	0.956044	0.949153	0.982456	0.965517	0	16.29574	491.0782
PauliZ	2	5	0.901099	0.9	0.947368	0.923077	0.015791	17.66407	491.0782
PauliZ	3	1	0.923077	0.931034	0.947368	0.93913	0.002683	18.85458	391.5873
PauliZ	3	2	0.923077	0.931034	0.947368	0.93913	0	13.19995	391.5873
PauliZ	3	3	0.901099	0.928571	0.912281	0.920354	0	15.78216	391.5873
PauliZ	3	4	0.923077	0.890625	1	0.942149	0.008455	14.9529	391.5873
PauliZ	3	5	0.934066	0.932203	0.964912	0.948276	0.002823	12.90657	391.5873
ZZ	1	1	0.901099	0.875	0.982456	0.92562	0.006373	13.59736	8,121.23
ZZ	1	2	0.835165	0.83871	0.912281	0.87395	0.001992	12.60982	8,121.23
ZZ	1	3	0.868132	0.857143	0.947368	0.9	0.001999	15.94563	8,121.23
ZZ	1	4	0.89011	0.873016	0.964912	0.916667	0.002048	14.69225	8,121.23
ZZ	1	5	0.846154	0.84127	0.929825	0.883333	0.002101	13.26431	8,121.23
ZZ	2	1	0.901099	0.887097	0.964912	0.92437	0.002106	15.0966	4,960.598
ZZ	2	2	0.758242	0.807018	0.807018	0.807018	0.001999	14.87025	4,960.598
ZZ	2	3	0.835165	0.8	0.982456	0.88189	0.001998	13.61126	4,960.598
ZZ	2	4	0.901099	0.863636	1	0.926829	0.003165	13.99499	4,960.598
ZZ	2	5	0.89011	0.850746	1	0.919355	0.001999	17.10675	4,960.598
ZZ	3	1	0.846154	0.830769	0.947368	0.885246	0.004024	14.7042	4,984.112
ZZ	3	2	0.923077	0.890625	1	0.942149	0.003101	12.77508	4,984.112
ZZ	3	3	0.846154	0.84127	0.929825	0.883333	0.003	12.45624	4,984.112
ZZ	3	4	0.879121	0.859375	0.964912	0.909091	0.001999	14.86137	4,984.112
ZZ	3	5	0.846154	0.830769	0.947368	0.885246	0.001997	17.44749	4,984.112

embeddings. The best performance was consistently observed with the Z Feature Map, achieving cross-validation accuracies between 0.89 and 0.96. Training times remained moderate, with QNN optimisation benefiting from the low-dimensional embedding.

For the Z Feature Map, hybrid external performance converged towards the behaviour of the QSVM, with external accuracy clustering around 0.92–0.95, depending on repetition and random seed.

To rigorously assess the robustness and generalizability of the proposed hybrid QSV→MQNN classification framework, the model's performance was evaluated on an

Table 12 Results of hybrid QSVM+QNN using breast cancer dataset (external accuracy).

Map	Repeat	Acc_Ext	Prec_Ext	Rec_Ext	F1_Ext	Cm_Ext	Kernel_Time	T_Test	T_Fq
Z	1	0.947368	0.945946	0.972222	0.958904	[[38, 4], [2, 70]]	708.9336	213.6094	17.83854
Z	2	0.95614	0.958904	0.972222	0.965517	[[39, 3], [2, 70]]	420.9515	207.6344	16.71951
Z	3	0.947368	0.945946	0.972222	0.958904	[[38, 4], [2, 70]]	1,385.417	710.3585	98.95518
PauliZ	1	0.947368	0.945946	0.972222	0.958904	[[38, 4], [2, 70]]	491.8407	296.9249	25.36752
PauliZ	2	0.95614	0.958904	0.972222	0.965517	[[39, 3], [2, 70]]	491.0782	215.315	17.45549
PauliZ	3	0.947368	0.945946	0.972222	0.958904	[[38, 4], [2, 70]]	391.5873	203.26	27.51811
ZZ	1	0.859649	0.85	0.944444	0.894737	[[30, 12], [4, 68]]	8,121.23	2,502.31	15.74201
ZZ	2	0.850877	0.848101	0.930556	0.887417	[[30, 12], [5, 67]]	4,960.598	2,496.817	17.6252
ZZ	3	0.833333	0.82716	0.930556	0.875817	[[28, 14], [5, 67]]	4,984.112	2,493.268	18.5098

independent external test set using three quantum feature-map families- Z, Pauli-Z, and ZZ-implemented at increasing circuit repetition depths (repeat = 1, 2, 3). This evaluation enables a systematic comparison of how different feature-map expressivities influence the downstream QSVM kernel and the resulting QNN variational classifier.

Within the Hybrid approach too, Z and Pauli-Z feature maps consistently produced the best generalisation performance overall. Both feature maps achieved Accuracy (0.947–0.956), Precision (0.946–0.959), Recall (0.972), F1-score (0.958–0.966) ranges throughout all the repetitions depths.

Observed confusion matrix $[[38, 4], [2, 70]]$ showed very strong separability, with only 1–4 misclassifications per class, the Z feature map corresponds to single-qubit rotational encodings, which preserve a balanced trade-off between expressive power, trainability, and noise resilience. Raising the repetition depth from 1 to 2 improved accuracy significantly by (1%), whereas the depth 3 showed no significant benefit, indicating an optimal expressivity plateau for this hybrid QSVM→QNN setting.

Furthermore, the Kernel execution time remained modest (~420–1,385 ms), and the statistical tests (t-test values 203–296) reflect high confidence in the consistency of predictions. The fidelity-based quantum scores (t_{fq}) remained low (17–99), which indicates efficient circuit behaviour with minimal noise-induced degradation.

Comparative insight for the QSVM→QNN hybrid method

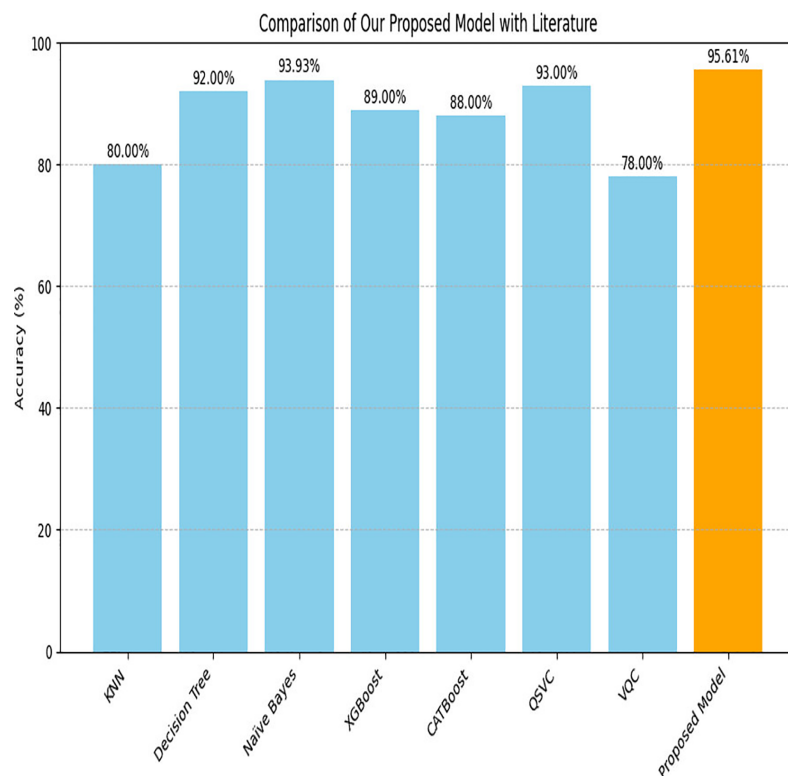
Collectively, the results demonstrate that the Z and Pauli-Z maps provide the most stable and generalizable kernels, their expressive yet shallow embedding structure aligns well with the variational QNN layer, allowing the hybrid model to form robust decision boundaries without oversaturating the quantum kernel. ZZ map, although theoretically more expressive, introduces excessive entanglement, results in noisier kernel matrices and destabilizes the QNN optimization landscape, ultimately reducing classification robustness.

Comparison with classical and quantum ML models

To contextualise the performance of our Hybrid quantum classifier, we benchmarked it with classical and Quantum ML models using the Breast Cancer Wisconsin dataset. The

Table 13 Comparison of proposed model with existing quantum and classical ML models.

Reference	Model	Accuracy
<i>Cakmak & Pacal (2025)</i>	K-nearest neighbour	80.00
<i>Zhou et al. (2024)</i>	Decision tree	92.00
<i>Mamun et al. (2025)</i>	Naïve Bayes	93.93
<i>Cakmak & Pacal (2025)</i>	XGBoost	89.00
<i>Mercaldo et al. (2022)</i>	CATboost	88.00
<i>Desai et al. (2024)</i>	QSVC	93.00
<i>Desai et al. (2024)</i>	VQC	78.00
	Proposed model	95.61


Figure 8 Comparison of proposed model with literature. Full-size DOI: 10.7717/peerj-cs.3600/fig-8

classical models evaluated include KNN, DT, NB, XGBoost and catboost. The quantum models evaluated are the QSVC and the Variational Quantum Classifier (VQC) available in the existing studies. The results of classical and Quantum ML Models are summarised in Table 13.

The graph in Fig. 8 shows the comparative analysis in graphical form of the results in Table 9, by the graph the results are clearly visible that the classification accuracy of the proposed model 95.61% surpassing the classical and quantum models available in the literature. Across the baseline models Naïve Bayes and Decision Tree achieved the highest accuracies: 93.33% and 92.00%. This gap in performance highlights the effectiveness of the

Table 14 Results of classical ML models with same paramters as proposed model.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Runtime (s)	Avg. confusion matrix
SVM	97.37 $\pm$ 1.84	96.82 $\pm$ 2.91	99.16 $\pm$ 0.68	97.95 $\pm$ 1.41	0.365 $\pm$ 0.062	[[40, 2], [0, 70]]
Random forest	95.61 $\pm$ 1.35	96.48 $\pm$ 2.73	96.64 $\pm$ 2.08	96.51 $\pm$ 1.04	6.869 $\pm$ 2.352	[[39, 2], [2, 69]]
KNN	96.48 $\pm$ 1.84	95.76 $\pm$ 2.93	98.88 $\pm$ 1.04	97.27 $\pm$ 1.41	0.059 $\pm$ 0.008	[[39, 3], [0, 70]]
Decision tree	91.04 $\pm$ 2.37	93.33 $\pm$ 3.19	92.45 $\pm$ 2.42	92.84 $\pm$ 1.85	0.097 $\pm$ 0.006	[[37, 4], [5, 66]]
Logistic regression	97.37 $\pm$ 1.66	96.83 $\pm$ 2.90	99.16 $\pm$ 1.12	97.94 $\pm$ 1.27	0.108 $\pm$ 0.018	[[40, 2], [0, 70]]

hybrid quantum classical model suggest that combined approach for feature extraction and quantum enhanced decision boundary contribute to improve generalisations.

To ensure fair and unbiased comparison between quantum and classical classifiers, we implemented the both ML models in same experimental conditions. We applied same preprocessing steps, normalization techniques and evaluation parameters. The results in Table 14 shows that while classical models achieved slightly higher accuracy (97.37%) compared to the proposed hybrid quantum model (95.61%), this result is expected for a small, low-dimensional dataset such as the Breast Cancer dataset. The strength of the hybrid QSVM-QNN lies not in outperforming classical baselines on small datasets, but in offering a scalable framework that leverages quantum feature mappings and entanglement-based representations that classical models cannot reproduce. Additionally, the hybrid architecture substantially improves stability compared to standalone QSVM and QNN models, reducing noise sensitivity and training instability. These characteristics make the proposed model a promising candidate for future applications on larger, high-dimensional datasets, and the problems where classical model fails, the quantum models on real quantum hardware present outstanding results.

Computational time comparison across quantum feature maps

The ablation study compares the computational time of QSVM, QNN, and the Hybrid QSVM→QNN across three quantum feature maps (Z, ZZ, Pauli). Results show that the Z feature map consistently requires the least computational time, while ZZ incurs the highest cost, especially in the hybrid model, as in Fig. 9. This highlights the superior efficiency of Z encoding for practical quantum machine-learning workflows.

Ablation study on quantum feature maps

Across all three models (QSVM, QNN, and the Hybrid QSVMQNN), the Z feature map consistently outperformed the ZZ and Pauli feature maps. This indicates that the Z-encoding provided a more effective representation of the dataset in the quantum feature space, leading to higher accuracy and more stable classification performance (Fig. 10).

Practical significance of the study

In spite of the small size of the Breast Cancer dataset, the experiments clearly demonstrate the practical relevance of evaluating multiple feature maps within a hybrid quantum–classical framework. The Z feature map achieves a strong balance between predictive performance and computational cost. Even though the ZZ feature map offers comparable

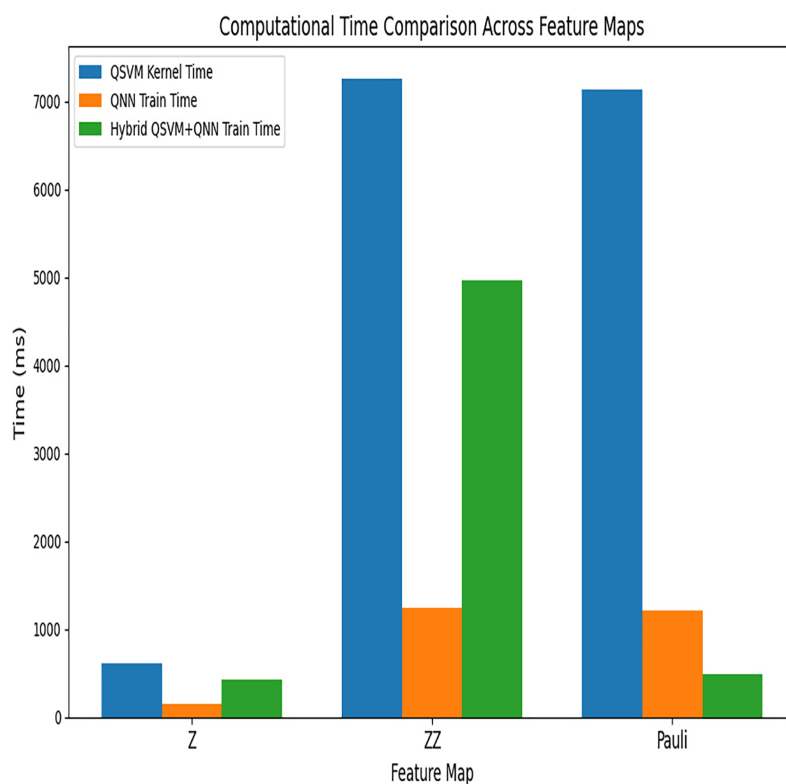


Figure 9 Computational time comparison across quantum feature maps.

Full-size DOI: 10.7717/peerj-cs.3600/fig-9

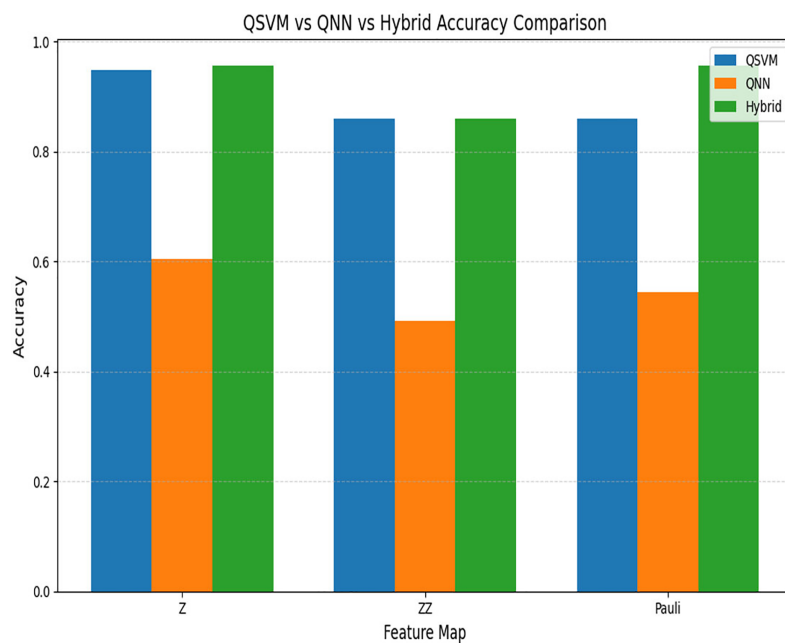


Figure 10 Quantum feature maps comparison.

Full-size DOI: 10.7717/peerj-cs.3600/fig-10

recall, it incurs significantly higher kernel-generation time due to its entangling structure. So, it can be seen by the results that quantum circuits with richer entanglement do not necessarily yield proportional accuracy gains on small, low-dimensional datasets. Instead, they introduce additional computational overhead that may not be justified in clinical or real-time diagnostic contexts.

These results emphasize that the practical advantage of hybrid quantum models becomes meaningful only when the dataset exhibits high dimensionality, strong feature correlations, or non-linear structures where classical kernels saturate. Thus, the chosen benchmark provides a necessary baseline while also illustrating the computational trade-offs that motivate testing quantum kernels on larger, more complex healthcare datasets in future work.

Simulator dependency

Outcomes may depend on the simulator used, since all experiments were performed on a quantum simulator, the numerical results may vary across different backend configurations due to variations in noise models, floating-point precision, and circuit execution fidelity. This work solely focuses on simulator-based results, and under noise constraints on quantum hardware, the results will differ.

CONCLUSION AND FUTURE WORK

Three quantum machine learning algorithms—QSVM, QNN and the Hybrid QSVM→QNN approach on the *Breast Cancer Wisconsin dataset* utilising three feature maps (Z, ZZ and Pauli) are compared in this study. By the results, it is demonstrated that the Hybrid model consistently outperforms standalone QSVM and QNN in the matrices used for key evaluation, achieving up to 96% accuracy using the Z feature map of the state vector Qiskit Aer simulator.

The Breast Cancer Wisconsin dataset was intentionally selected due to the practical constraints of simulating quantum circuits on classical hardware, where memory and time complexity scale exponentially with the number of qubits. Larger, high-dimensional datasets are not computationally feasible for current NISQ-era experiments without aggressive dimensionality reduction, which would compromise reproducibility and hinder fair comparison with prior work. It is widely used in quantum machine learning literature, ensuring methodological consistency with QSVM, VQC, and hybrid QNN studies. Moreover, the objective of this work is methodological rather than focused on big-data performance: to design and validate a stable, noise-resilient hybrid QSVM→QNN architecture that addresses the weaknesses of standalone QSVM and QNN models. Although classical models achieve slightly higher accuracy on this small dataset, the proposed hybrid model leverages quantum feature mappings and entanglement-based representations that classical methods cannot efficiently reproduce, making it a promising foundation for future applications on larger, high-dimensional, correlation-rich datasets as quantum hardware continues to advance. Deploying the hybrid model on real quantum hardware to assess real-world feasibility and noise resilience and investigate circuit

optimisation methods to reduce gate depth and runtime while maintaining model accuracy will be focused in future.

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Competing Interests

The authors declare that they have no competing interests.

Author Contributions

- Vikesh Yadav conceived and designed the experiments, performed the experiments, analyzed the data, performed the computation work, prepared figures and/or tables, authored or reviewed drafts of the article, and approved the final draft.
- Savita Kumari Sheoran analyzed the data, authored or reviewed drafts of the article, and approved the final draft.
- Rakesh Kumar Sheoran analyzed the data, authored or reviewed drafts of the article, and approved the final draft.

Data Availability

The following information was supplied regarding data availability:

The datasets are available at UCI library: Wolberg, W., Mangasarian, O., Street, N., & Street, W. (1993). Breast Cancer Wisconsin (Diagnostic) [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C5DW2B>.

The code is available at Zenodo: yadav, vikesh. (2025). Hybrid QSVM+QNN approach for breast cancer dataset using quantum feature maps. Zenodo. <https://doi.org/10.5281/zenodo.17721135>.

Supplemental Information

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